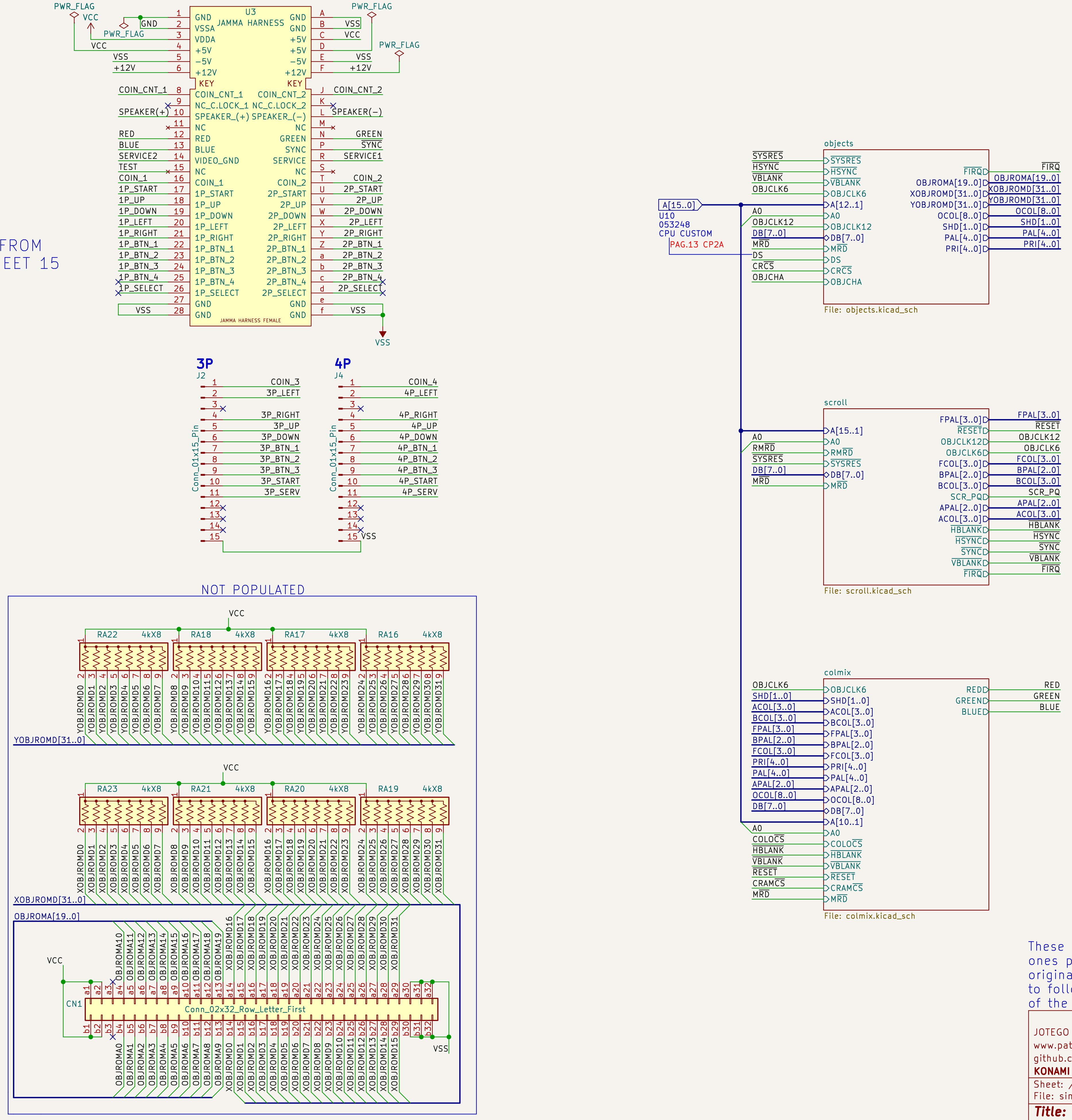

PWR_FLAG
VCC
GND
VSS
+12V
U3
JAMMA HARNESS
JAMMA HARNESS FEMALE
GND
VSSA
VDDA
+5V
-5V
+12V
KEY
COIN_CNT_1
COIN_CNT_2
NC_C.LOCK_1
NC_C.LOCK_2
SPEAKER_(+)
SPEAKER_(-)
SPEAKER(+)
SPEAKER(-)
NC
RED
GREEN
BLUE
SYNC
VIDEO_GND
SERVICE
SERVICE1
SERVICE2
TEST
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
3P
J2
Conn_01x15_Pin
COIN_3
3P_LEFT
3P_RIGHT
3P_UP
3P_DOWN
3P_BTN_1
3P_BTN_2
3P_BTN_3
3P_START
3P_SERV
4P
J4
COIN_4
4P_LEFT
4P_RIGHT
4P_UP
4P_DOWN
4P_BTN_1
4P_BTN_2
4P_BTN_3
4P_START
4P_SERV
NOT POPULATED
RA22 4kX8
RA18 4kX8
RA17 4kX8
RA16 4kX8
RA23 4kX8
RA21 4kX8
RA20 4kX8
RA19 4kX8
YOBJROMD[31..0]
XOBJROMD[31..0]
OBJROMA[19..0]
CN1
Conn_02x32_Row_Letter_First
A[15..0]
U10
053248
CPU CUSTOM
PAG.13 CP2A
objects
SYSRES
HSYNC
VBLANK
OBJCLK6
A[12..1]
A0
OBJCLK12
DB[7..0]
MRD
DS
CRCS
OBJCHA
FIRQ
OBJROMA[19..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
OCOL[8..0]
SHD[1..0]
PAL[4..0]
PRI[4..0]
File: objects.kicad_sch
scroll
A[15..1]
RMRD
FPAL[3..0]
RESET
OBJCLK12
OBJCLK6
FCOL[3..0]
BPAL[2..0]
BCOL[3..0]
SCR_PQ
APAL[2..0]
ACOL[3..0]
HBLANK
HSYNC
SYNC
VBLANK
FIRQ
File: scroll.kicad_sch
colmix
A[10..1]
COLOCS
CRAMCS
RED
GREEN
BLUE
File: colmix.kicad_sch

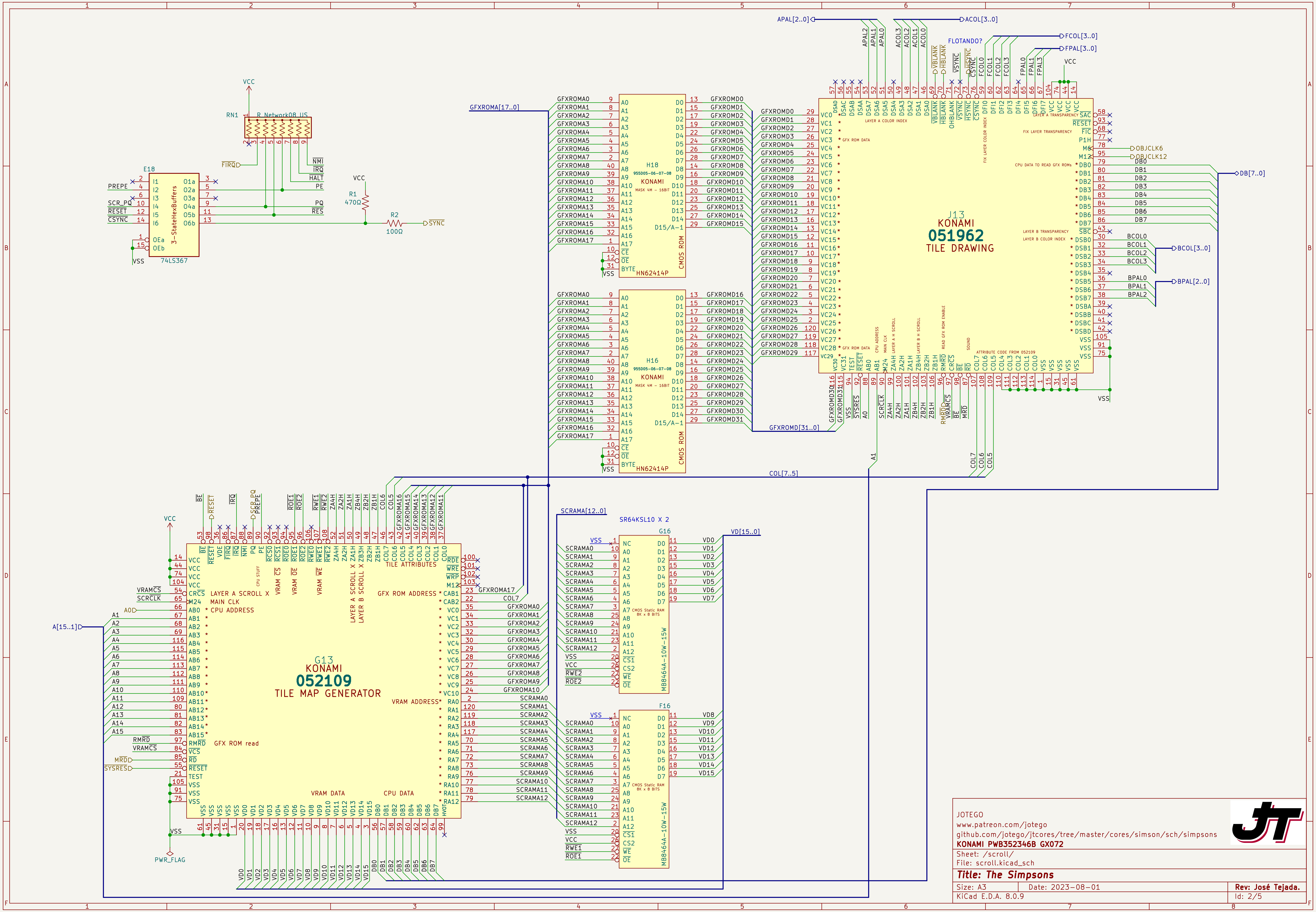
THESE SIGNALS COME FROM
SCH SIMPSONS I/O SHEET 15

These schematics cover the missing pages in the original ones published by the maker. Signal names existing in the original pages have been preserved. New signal names try to follow the maker style or are taken from similar schematics of the same maker.

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB352346B GX072
Sheet: /
File: simpsons.kicad_sch

Title: The Simpsons

Size: A3
Date: 2023-08-01
Rev: José Tejada.
KiCad E.D.A. 8.0.9
Id: 1/5


E18
74LS367
3-StateHexBuffers
PREPE
SCR_PQ
RESET
CSYNC
VSS
RN1
R_Network08_US
VCC
FIRQ
NMI
IRQ
HALT
PE
PQ
RES
R1
470Ω
R2
100Ω
SYNC
GFXROMA[17..0]
H18
H16
955005-06-07-08
KONAMI
MASK 4M - 16BIT
CMOS ROM
HN62414P
GFXROMD[31..0]
APAL[2..0]
ACOL[3..0]
FLOTANDO?
FCOL[3..0]
FPAL[3..0]
J13
KONAMI
051962
TILE DRAWING
LAYER A COLOR INDEX
GFX ROM DATA
FIX LAYER TRANSPARENCY
LAYER A TRANSPARENCY
CPU DATA TO READ GFX ROMs
LAYER B TRANSPARENCY
LAYER B COLOR INDEX
ATTRIBUTE CODE FROM 052109
OBJCLK6
OBJCLK12
DB[7..0]
BCOL[3..0]
BPAL[2..0]
COL[7..5]
A1
SCRAMA[12..0]
SR64KSL10 X 2
G16
F16
CMOS Static RAM 8K x 8 BITS
MB8464A-10W-15W
VD[15..0]
RWE2
ROE2
RWE1
ROE1
G13
KONAMI
052109
TILE MAP GENERATOR
LAYER A SCROLL X
MAIN CLK
CPU ADDRESS
LAYER A SCROLL X
LAYER B SCROLL X
TILE ATTRIBUTES
GFX ROM ADDRESS
VRAM ADDRESS
GFX ROM read
VRAM DATA
CPU DATA
VRAMCS
SCRCLK
A0
A[15..1]
RMRD
MRD
SYSRES
PWR_FLAG
GFXROMA17
COL7

JOTEGO
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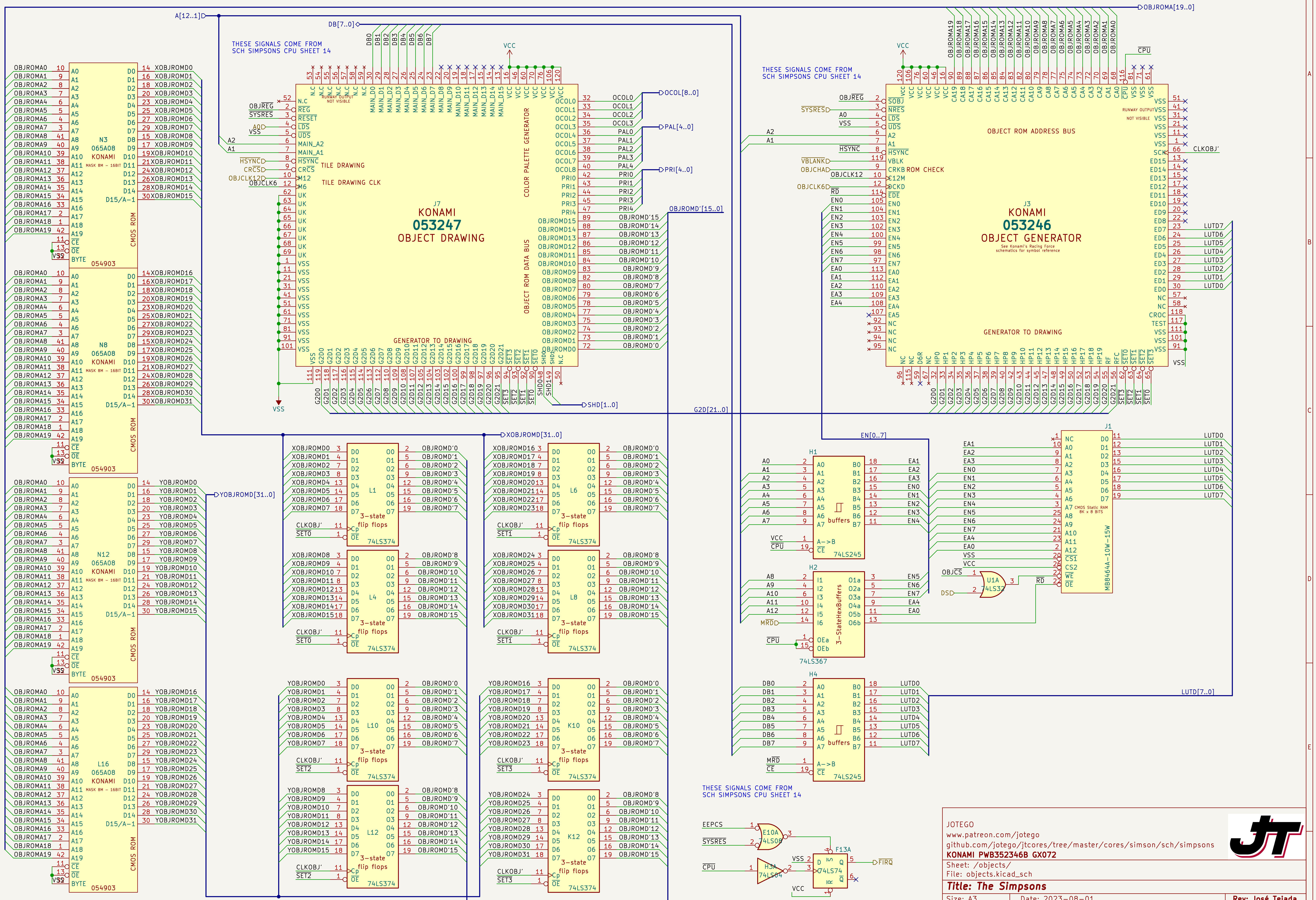
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB352346B GX072
Sheet: /scroll/
File: scroll.kicad_sch

Title: The Simpsons

Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 2/5

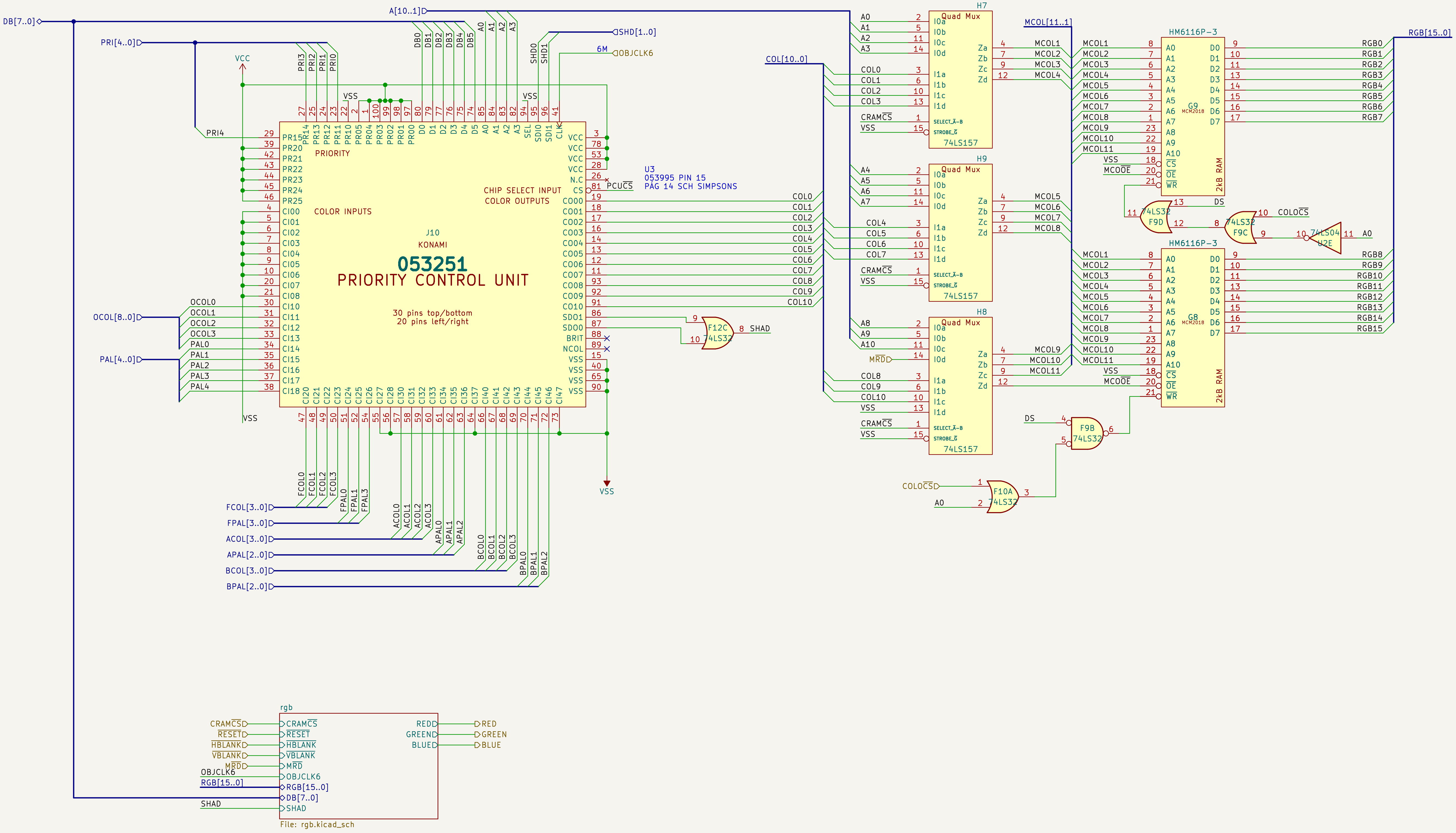

A[12..1]
DB[7..0]
OBJROMA[19..0]
THESE SIGNALS COME FROM
SCH SIMPSONS CPU SHEET 14
VCC
VSS
N3
065A08
KONAMI
MASK 8M - 16BIT
CMOS ROM
D15/A-1
BYTE
054903
N8
N12
L16
J7
KONAMI
053247
OBJECT DRAWING
RUNWAY OUTPUT NOT VISIBLE
TILE DRAWING
TILE DRAWING CLK
COLOR PALETTE GENERATOR
OBJECT ROM DATA BUS
GENERATOR TO DRAWING
OCOL[8..0]
PAL[4..0]
PRI[4..0]
OBJROMD'[15..0]
SHD[1..0]
G2D[21..0]
XOBJROMD[31..0]
YOBJROMD[31..0]
J3
KONAMI
053246
OBJECT GENERATOR
See Konami's Racing Force schematics for symbol reference
OBJECT ROM ADDRESS BUS
ROM CHECK
CLKOBJ'
EN[0..7]
LUTD[7..0]
L1
L4
L6
L8
L10
L12
K10
K12
3-state flip flops
74LS374
H1
H4
buffers
74LS245
H2
3-StateHexBuffers
74LS367
J1
CMOS Static RAM 8K x 8 BITS
MB8464A-10W-15W
U1A
74LS32
E10A
74LS08
H3A
74LS04
F13A
74LS74
EEPCS
SYSRES
CPU
FIRQ
OBJCS
DSD
RD
MRD
CE

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB352346B GX072

Sheet: /objects/
File: objects.kicad_sch

Title: The Simpsons

Size: A3 Date: 2023-08-01 Rev: José Tejada.
KiCad E.D.A. 8.0.9 Id: 3/5

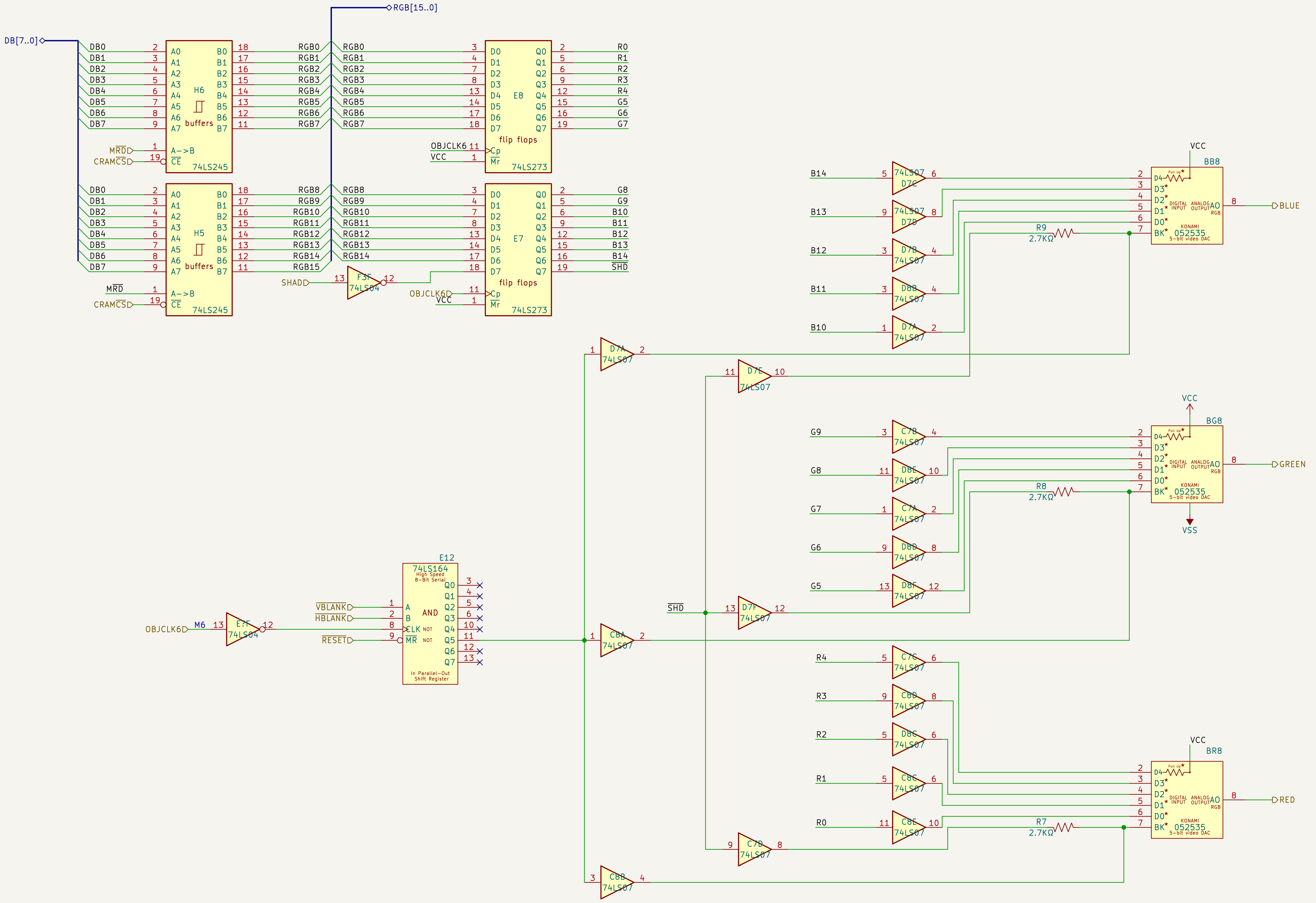

DB[7..0]
A[10..1]
PRI[4..0]
SHD[1..0]
OBJCLK6
6M
VCC
VSS
PRI4
J10
KONAMI
053251
PRIORITY CONTROL UNIT
30 pins top/bottom
20 pins left/right
PRIORITY
COLOR INPUTS
CHIP SELECT INPUT
COLOR OUTPUTS
U3
053995 PIN 15
PÁG 14 SCH SIMPSONS
PCUCS
OCOL[8..0]
PAL[4..0]
FCOL[3..0]
FPAL[3..0]
ACOL[3..0]
APAL[2..0]
BCOL[3..0]
BPAL[2..0]
F12C
74LS32
SHAD
COL[10..0]
MCOL[11..1]
RGB[15..0]
H7
H9
H8
Quad Mux
74LS157
SELECT_Ā-B
STROBE_Ḡ
CRAMCS
MRD
HM6116P-3
G9
G8
MCM2018
2kB RAM
MCOE
DS
F9D
F9C
F9B
F10A
74LS32
U2E
74LS04
COLOCS
A0
rgb
CRAMCS
RESET
HBLANK
VBLANK
MRD
OBJCLK6
RGB[15..0]
DB[7..0]
SHAD
REDD
GREEND
BLUED
RED
GREEN
BLUE
File: rgb.kicad_sch

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB352346B GX072

Sheet: /colmix/
File: colmix.kicad_sch

Title: The Simpsons

Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 4/5


DB[7..0]
RGB[15..0]
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
H6
buffers
74LS245
H5
MRD
CRAMCS
A->B
CE
RGB0
RGB1
RGB2
RGB3
RGB4
RGB5
RGB6
RGB7
RGB8
RGB9
RGB10
RGB11
RGB12
RGB13
RGB14
RGB15
E8
E7
flip flops
74LS273
OBJCLK6
VCC
Cp
Mr
SHADD
F3F
74LS04
R0
R1
R2
R3
R4
G5
G6
G7
G8
G9
B10
B11
B12
B13
B14
SHD
74LS07
D7C
D7B
D8B
D7A
D7E
D7F
D8E
C7A
D8D
D8F
C8A
C7C
C8D
D8C
C8C
C8E
C7B
C8B
C7B
BB8
BG8
BR8
052535
KONAMI
5-bit video DAC
DIGITAL INPUT
ANALOG OUTPUT
BK
AO
RGB
R9
R8
R7
2.7KΩ
BLUE
GREEN
RED
VSS
E12
74LS164
High Speed 8-Bit Serial
In Parallel-Out Shift Register
AND
VBLANK
HBLANK
RESET
M6
E7F
OBJCLK6
CLK
MR
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7

JOTEGO
www.patreon.com/jotego
github.com/jotego/jtcores/tree/master/cores/simson/sch/simpsons
KONAMI PWB352346B GX072

Sheet: /colmix/rgb/
File: rgb.kicad_sch

Title: The Simpsons

Size: A3 Date: 2023-08-01 **Rev: José Tejada.**
KiCad E.D.A. 8.0.9 Id: 5/5